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Alternative liquid fuels in the UK in the inter-war period (1918–1938): Insights from a failed energy transition

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ABSTRACT

Against a backdrop of growing interest in low carbon alternatives to petroleum derived liquid fuels, this paper analyses two early attempts to develop alternative liquid fuels to rival petrol, the incumbent fuel. We focus on two alternatives manufactured in the UK during the inter-war period (1918–1938), a power alcohol and a petrol-from-coal. Both fuels received government support during a time of rapid growth in the motor industry, fluctuating economic conditions and fears of absolute oil shortages. Governance of fuel distribution in particular had significant effects on the economic feasibility of both fuels and their ability to penetrate a market dominated by the oil industry incumbents. The paper presents a rich, complex story of how the development of these fuels interacted with fluctuating political perceptions of energy security, changes in the framing of energy and shifts in governance, in what were to become failed attempts at path creation and transition.

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1. Introduction

Experimentation with alternatives to liquid fuels derived from petroleum was commonplace during the first decades of the 20th century. Pressures to develop alternatives to crude oil derivatives, however, reached a climax in the inter-war period, particularly among countries without indigenous petroleum resources, including Britain (Fig. 1: Egloff, 1938). These alternatives were rarely economically competitive with petroleum-based fuels, and required various forms of state support (Committee on Industrial Alcohol, 1905; Falmouth Committee, 1938; San Román and Sudriá, 2003).

In this paper, we present two empirical case studies that examine the evolution, state support for, and commercial failure of two liquid fuels developed in Britain during the inter-war period (1918–1938). Building on Bennett (2009) and Bennett and Pearson (2009), we focus on power alcohol produced by the Distillers Company Ltd. (DCL) and petrol derived from coal by Imperial Chemical Industries Ltd. (ICI).

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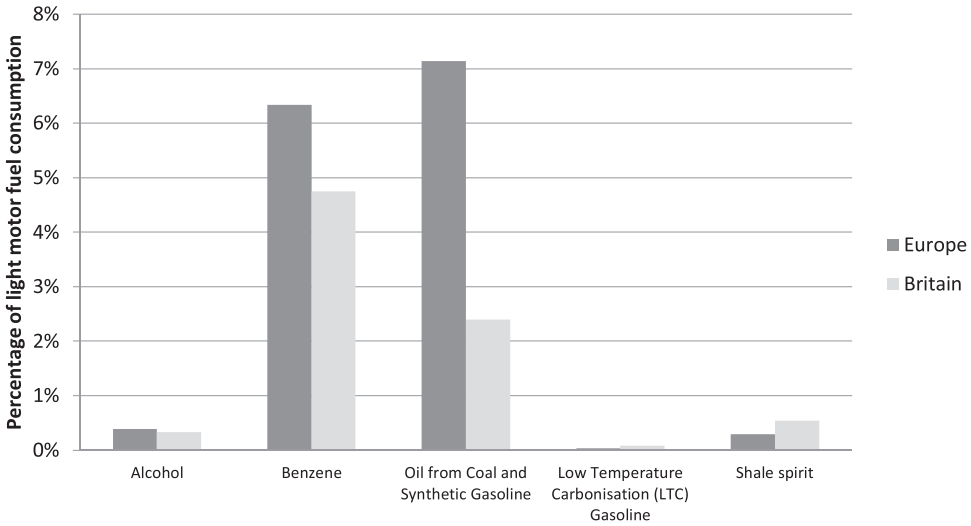

Fig. 1. European and British synthetic fuel consumption as a percentage of total light motor fuel consumption, 1937.
Source: Egloff (1938). Countries included: Germany, Estonia, Czechoslovakia, Lithuania, Hungary, Poland, Latvia, Yugoslavia, Belgium, France, United Kingdom, Austria, Sweden, Holland, Finland and Switzerland.

Empirically, our interest in past attempts to develop alternatives to mainstream transport fuels stems from the enduring concern with their development, for reasons that include energy security, desires to support feedstock industries (from coal to agriculture), and now their role in a low-carbon transition (Office for Low Emission Vehicles, 2013). Both fuels discussed here received direct and indirect state support during the inter-war years and were chosen partly for this reason, because support for the introduction of new fuels and technologies is currently a politically salient issue in the context of sustainability transitions. Here, direct support refers to state subsidies, whilst indirect support relates to support for research, development and demonstration (RD&D). Although other alternatives were developed during this time (Fig. 1), their production was more diffuse and is not examined further here.

While extensive business histories of DCL and ICI have been written elsewhere (e.g. Weir, 1995 for DCL; Reader, 1975, 1977 for ICI), our approach differs from these analyses. Specifically, we are interested in the processes which led to the emergence of these two fuels, the growth and subsequent removal of policy support for them, and to their eventual failure to challenge the incumbent petroleum industry. By building on a governance logics action-space framework (Foxon et al., 2010; Foxon 2013) we explore how changes in energy security led to structural shifts in governance logics, and show how these changes influenced legitimacy for and the consistency of policy support. In addition, as explained in the next section, we adopt a relational approach (Garud and Gehman, 2012; Jørgensen, 2012; Longhurst and Chilvers, 2013) to understand how actors implicated in the emergence of these two fuels framed the development of these fuels to meet their own ends.

Empirical case studies of long-term historical transitions have enhanced understanding of the dynamics of transition and persistence, and have been helpful in testing and developing theories of change and informing policy and practice (Geels, 2002; Grin et al., 2010; Fouquet and Pearson, 2012; Garud and Gehman, 2012; Watson et al., 2012; Smith, 2013). However, while much attention has been given to past transition successes and some to the potential for incumbents to engage with 'path creation' and 'creative accumulation' (Bergek et al., 2013), more attention is gradually and rightly being paid to their roles and strategies in transition failure and resisting change (Magnusson and Berggren, 2011; Wells and Nieuwenhuis, 2012; Geels, 2014; Mazur et al., 2015). Our two case studies each concern one industry, one dominant firm in that industry, one technology and one fuel. Both firms sought and failed to gain permanent entry into a market dominated by the incumbents and, therefore, offer an unusual opportunity to gain insights into state-industry relationships, the causes and dynamics of transition failures and the role of incumbents in them.

Historical case studies can also illustrate the possibility of radical transformation and challenge naïve perspectives or received wisdom about past successes and failures of sociotechnical transitions, policy interventions, and technologies (Tilly, 2006; Thane, 2009; Bayly et al., 2011; Hirsch and Jones, 2014). Watson et al. (2012) recognise that the quality of historical case studies as analogues lies not in their perfect 'fit' with contemporary technologies or circumstances, but in being similar in one aspect in particular. In light of this, empirically, we aim to provide a useful historical analogue for understanding the emergence of support for technological substitutes within socio-technical transitions. In particular, we draw attention to how our case studies provide insights into the support for and development of contemporary alternative fuels and the power of incumbents, particularly in terms of ownership of specialised complementary assets such as fuel distribution infrastructure (Teece, 1986). Conceptually, we aim to develop an understanding of the dynamic relationship between energy security, the framing of energy, governance logics and the policy environment.

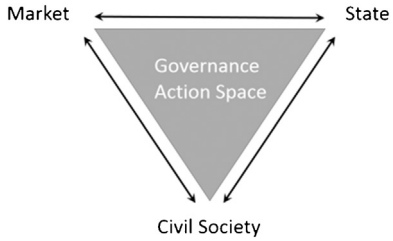

Fig. 2. The governance logics action space framework (adapted from Foxon et al., 2010).

The paper is structured as follows: Section 2 describes the analytical approach; Section 3 discusses the development of the fuels and the influence of the incumbent oil industry; Section 4 presents the results and discussion around the themes of state-industry interactions and support, energy security and hybrid governance; Section 5 concludes the paper and presents insights for sustainability transitions from this failed transition.

2. Analytical approach

We adopt two approaches to help examine the emergence and subsequent removal of policy support for two alternative liquid fuels that emerged in inter-war Britain, eventually disappearing completely from the market in the early post-World War II (WWII) years. First we describe an extended governance logics action-space approach in which the governance logics interact with changing perceptions of energy security and the framing of energy, with impacts on policy and regulation. This is used to broadly describe and explain structural shifts within and between governance logics, or 'rules of the game'; shifts that changed political legitimacy for state support of the two alternative liquid fuels. We then describe a relational approach employed both to examine how actors implicated in the emergence of these two fuels framed their development and tried to influence the policy environment, and to assess the degree to which their strategies succeeded.

Governance is implicit within the notion of intervention. Responding to criticisms of transition theories' lack of connection with governance (Smith et al., 2005) we recognise the need to integrate key actors, structures, and institutions into governance dynamics. To do this we employ an extension to the 'governance logics action space' framework first articulated by Jacquie Burgess, Tom Hargreaves and colleagues in the Transitions Pathways project (Foxon, 2013). This approach draws on the concept of institutional and organisational field logics from institutional theory, defined by Thornton and Ocasio (1999: 804) as: '*the socially constructed, historical patterns of cultural symbols and material practices, assumptions, values and beliefs by which individuals produce and reproduce their material subsistence, organise time and space, and provide meaning to their daily activity*'.

The governance logics action space framework has been used as a heuristic to help explore future energy transition pathways (Foxon et al., 2010) and the governance of contemporary energy systems (Foxon 2013; Bolton and Foxon, 2013). It envisions a transition pathway arising through the interactions of three broad and highly aggregated types of governance 'logics' (state, market, civil society) and the shifting balances of agency between them (Fig. 1). Fuenfschilling and Truffer (2014) identified a similar set of logics in their examination of the Australian water sector.

These fundamentally different logics influence the framing of key energy challenges and responses to them, including policy responses (Foxon, 2013), although they are not the only factors that influence the actual outcomes. During a transition, actors attempt to enrol others into their preferred governance logic, with the most successful actors defining the dominant logic within the action space (the shaded area in the centre of Fig. 2). Where a mixture of logics coexists (as at present in the UK's mix of market and state logics), this is termed 'hybrid governance' (Bolton and Foxon, 2013), or 'semi-coherence' (Fuenfschilling and Truffer, 2014). It should be noted, however, that in our approach, not only does the interaction between the logics in the action space vary with changing circumstances and actor agency, but also the logics themselves alter, both through this interaction and as views about the roles of state, market and civil society in energy respond to and reflect wider, sometimes medium or longer term socioeconomic, ideological and political developments (Pearson and Watson, 2011).

Changing perceptions of energy security can exercise significant influence on the interplay between governance logics. Energy resources are extracted, transformed, distributed and delivered to provide energy services, including transport. The availability of cheap, reliable energy services and the resources to deliver them has been central to the functioning of sociotechnical systems in both developed and emerging economies (Fouquet, 2008; Haas et al., 2008). While energy security is a multifaceted, frequently contested notion, one aspect—the desire for stable, uninterrupted and cheap supplies of fossil fuels, including oil, coal and gas, and their products—has played a significant role in post-WWII energy governance and policy in Britain (Mitchell et al., 2013). Here, we show this was also the case during the interwar period in relation to securing supplies of oil and products for the armed forces in case of war.

Grimston (2010) suggests that structural shifts in the framing of energy occur with shifts in perceptions of energy security. When energy is seen as insecure (e.g. uncertain availability or price volatility driven by geopolitical events), energy tends to be framed as a 'social service'. State interventions, including control of supply and distribution, then have a higher degree of political legitimacy. In contrast, during periods of perceived energy security, energy is considered as a 'market commodity'.

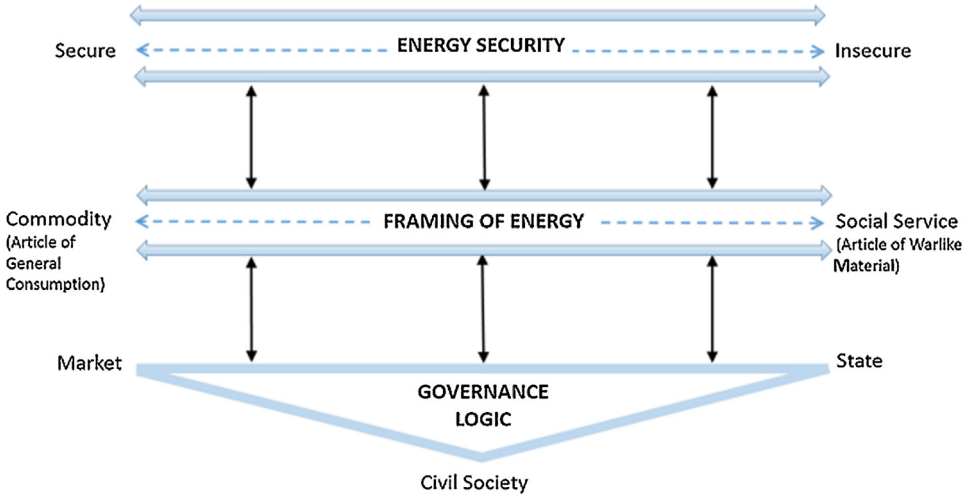

Fig. 3. The dynamic relationship between energy security, framing of energy, and governance logics.

State intervention in the marketplace is minimal beyond measures to prevent unfair competition and internalise economic externalities e.g. environmental pollution (Helm, 2005; Grimston, 2010; Fudge et al., 2011; Goldthau, 2012; Bolton and Foxon, 2013; Rydin et al., 2013). Thus, the dynamics of energy security affect the framing of energy as either a 'social service' (instability) or a 'market commodity' (stability).

Linking the dynamics in energy security and the framing of energy to the governance logics action space approach implies that during significant periods of energy insecurity, for example, a state logic tends to become more dominant. Conversely, when energy supplies are more secure, a market logic dominates. Fig. 3 provides a simplified illustration of how energy security, energy framing, and governance logics relate to and dynamically interact with one another *in extremis* and in more intermediate states.

Shifts in logic can have significant influences on current and future pathways, through their impacts on approaches to policy challenges, visions of future societies, and the shape of future energy systems. Our contention is that during the period of analysis covered in this paper, perceptions of energy security and concerns specifically related to the supply and cost of liquid fuels led to structural shifts in the balance between governance logics, with repercussions for the framing of energy, the perceived legitimacy of state intervention and the associated policies.

The extended governance logic action space approach described above provides a framework within which to understand how the dynamics of energy security stimulated structural shifts in logics, and subsequent perceived legitimacy for state intervention in relation to these fuels. To understand and explain how and why the particular policy instruments that supported two alternative liquid fuels emerged at a particular time, and were later withdrawn, we turn to a relational approach or 'insider ontology' (Garud et al., 2010; Garud and Gehman, 2012; Jørgensen, 2012; Smith and Raven, 2012; Longhurst and Chilvers, 2013). Building on science and technology studies (STS) (e.g. Callon, 1986; Latour, 1986; Law, 1991), a relational approach adopts an ontology that flattens distinctions in agency and structure, capturing the different roles of actors, the ways they frame issues, and the degree to which they are successful in enrolling other actors to their way of thinking.

A variety of conceptual, theoretical, and heuristic tools have been used to examine contemporary and historical socio-technical systems and their transformation (Garud and Gehman, 2012; Markard et al., 2012). Of these, the multi-level perspective (MLP) has been the most influential in empirical case-studies (Geels, 2002, 2012, 2014). While our analysis is usefully informed by the MLP, and although earlier criticisms that the MLP tended to underplay the mechanisms by which governance, politics, agency, and power influence transitions (Smith et al., 2005, 2010; Genus and Coles, 2008; Shove and Walker, 2010; Meadowcroft, 2011; Jørgensen, 2012) have been variously addressed (e.g. Geels, 2011, 2014), we find that for this study a relational approach addresses more directly the question of how the particular policy environment relating to our cases emerged. This is due to its focus on actor networks, co-construction and agency. We follow Smith and Raven (2012) in using this approach to understand how actors implicated in the emergence of these two fuels dynamically framed their development and actively attempted to shape the policy environment as they navigated the unfolding shifts in governance logics. As such, in this paper we bring agency, politics and power to the fore.

Spaces of potential intervention within large-scale infrastructure systems (energy, water, mobility) have been described in technological innovation systems (TIS) and transition management (TM) research (e.g. Jonsson, 2000; Frantzeskaki and Loorbach, 2010; Andersen, 2014). These specifically identify elements in infrastructure systems, assessing which need to change in order for the system as a whole to transition. However, TIS and TM approaches focus mainly on what is important in terms of infrastructure functionality rather than aspects that are important to actors with interests in the infrastructure,

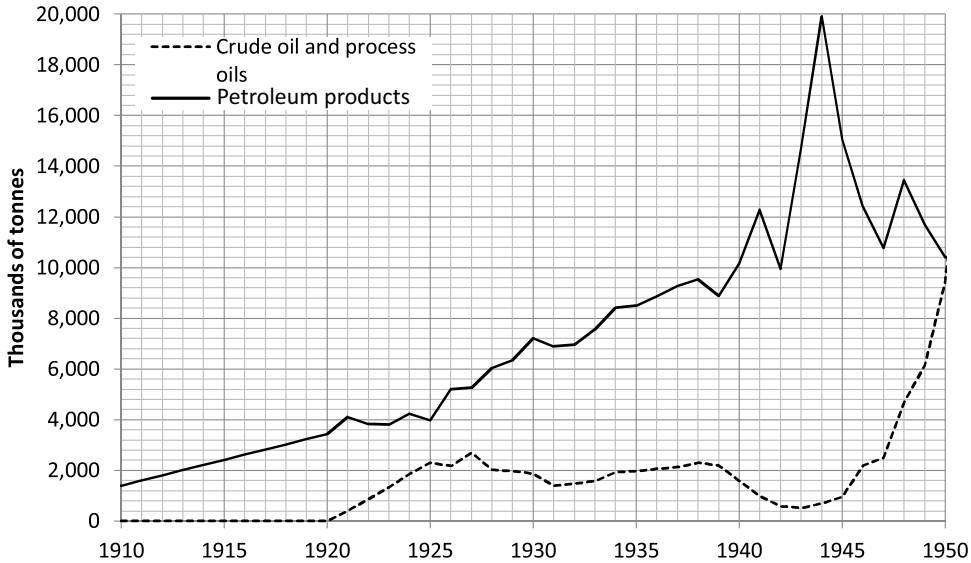

Fig. 4. UK Imports of crude oil and petroleum products, 1910–1950, in thousands of tonnes.

Source: DECC (2013)

including control over and access to that infrastructure. This is an issue that we show to be of considerable significance in the failed transition examined here.

Given this, we argue that a relational approach contributes to transitions research by shifting the focus to actor interests, using these to examine the scope for potential intervention in energy services, whilst recognising that spaces of intervention are contained within the structures and institutions of the energy infrastructure, including regulations and ownership of distribution infrastructures and the material structures themselves (Andersen, 2014).

We suggest that the governance logics action space framework and relational approaches help us to understand the tensions, choices, and decisions made by key actors involved in the development of alternative liquid fuels during a period of rapid macroeconomic, political, and social change. While the two approaches may appear to be at odds with one another ontologically, we have found this dual approach helpful in interpreting the processes leading to the eventual commercial failure of the two fuels in the 1950s.

Using original archival and other documentary material and secondary literature, we examine how relationships between industry and government, technological developments, the emerging fuel distribution infrastructure and prevailing macroeconomic conditions co-produced policy and market environments for alternative fuels. We see how, when combined, these interactions helped to shape governance logics, energy security and the framing of energy in ways that contributed ultimately to the demise of DCL's power alcohol and ICI's petrol-from-coal.

3. The development of the fuels and the influence of the incumbent oil industry

The economic and defence benefits of liquid fuels were increasingly recognised in Britain before World War I (WWI) (e.g. Royal Commission on Fuels and Engines, 1913). But it was only after WWI that imports of crude oil and other derivatives grew rapidly (Fig. 4). This was in response to increasing demand from the naval and marine mercantile sectors, and changes in civil society which saw rapid growth in motorcar ownership, road freight, and aviation (Fig. 5; Johnson et al., 2014).

Infrastructure for liquid fuel distribution also developed considerably during WWI, reducing fuel costs (Dixon, 1963). Coupled to the growing societal acceptance of liquid fuels (Podobnik, 2006) and spread of fuel-using technologies and their associated distribution infrastructure, by the 1920s a cultural narrative and vision developed that the 'oil age' represented significant technological progress and was unstoppable (e.g. Lawton, 1925; Cadman, 1931; The Institute of Fuel, 1936). Industrial action by the coal industry further created a window of opportunity for the emergence of liquid fuels in the transport and industrial sectors (see for example, Chelmsford Chronicle, 1921).

Throughout the inter-war period, these technological and fuel changes in the transport sector significantly influenced, and were influenced by, government strategies to ensure security of liquid fuel supply. Severe economic instability during the inter-war period, including the depressions of 1919–21 and 1930–31 (Broadberry, 1986), rising demand for imported petroleum, widening trade deficits and a growing trend of economic nationalism across Europe, partly in response to growing public debt, highlighted the perceived value of domestic production of alternative liquid fuels. Furthermore, WWI had emphasised the strategic importance of liquid fuels and the consequences of heavy reliance on international supply chains.

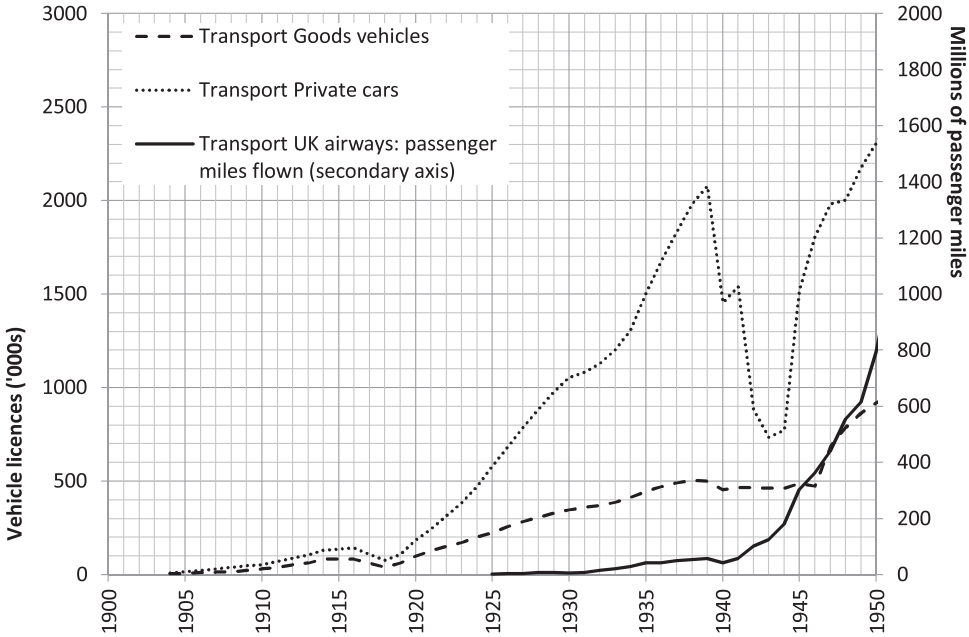

Fig. 5. Growth in motor vehicle use for goods vehicles and private cars in thousands of vehicle licences (primary axis) and growth in civil aviation in millions of passenger miles (secondary axis), 1897–1960.
Data sources: 1928–1960 from Society of Motor Manufacturers and Traders (1973) in Pugh (2008); 1914 from Pugh (2008); 1897–1905 from Wood (1996).

In the next three sub-sections, we briefly outline the history of the two fuels, the differences between the case studies and the position and influence of the oil majors. More detailed information about the case studies can be found in Johnson et al. (2014).

3.1. Power alcohol and Distillers Company Limited (DCL)

Power alcohol was used as motor fuel, either on its own or as part of a blend (Weir, 1995). It differed from petrol in that it possessed 'anti-knock' qualities due to its high octane level, making it suitable for high-compression engines (often found in commercial vehicles, sports, and motor-racing cars). It was said to offer improved engine power, increased mileage, and lower running costs than petrol (Commercial Motor Archive, 1923).

Power alcohol was produced from molasses by DCL. Formed in 1877, DCL was an amalgamation of several Scottish whisky distilleries. It grew to become one of the largest distilling companies in the UK, and with it a powerful force in the UK's distilling industry. DCL also produced industrial alcohol, which was used extensively in the chemical industry. Although power alcohol differed from industrial alcohol both in terms of its chemical properties and uses, its fortunes were strongly influenced by decisions made by government on industrial alcohol. Domestic production of industrial alcohol was seen by many users as uncompetitive with imports from Germany and America (Committee, 1905). The high cost of domestic industrial alcohol was attributed to Excise restrictions originally imposed on potable alcohol, preventing simultaneous brewing and distilling and no brewing, fermentation, or distillation on a Sunday. It was argued that these restrictions limited output and contributed to the higher cost of industrial alcohol over imports. The government wanted to be seen as supporting home industry. As such, in 1906 DCL received a subsidy for its industrial alcohol, enabling its use in the chemical industry to increase substantially (Committee, 1905).

By 1913, rapid increases in car ownership, and growing fuel security concerns, led the government to convene a Committee to examine the potential for a home-produced fuel—power alcohol. Following the Committee's recommendations, and conscious of protecting home industry from competition from the increasingly powerful international oil industry (Second Report on Motor Fuel, 1921), power alcohol benefitted from the same tax exemptions available to other domestically produced fuels available at the time (benzol from coal and shale-oil) and the subsidy on industrial alcohol was extended to power alcohol (Committee, 1919). This made power alcohol cost competitive with high-grade petrol.

Consumption of power alcohol, and production of industrial alcohol, increased dramatically throughout the 1920s, leading the Treasury to question the cost of the subsidy (Marsden, 1943). No investigative steps were taken until 1944, however, when the May Committee was convened. The Committee found that because domestic production of large quantities of alcohol for munitions during wartime had led to the removal of Excise restrictions, the conditions under which power and industrial alcohol were produced (which had justified the subsidy) were no longer applicable (May Committee, 1945). This finding, combined with greater certainty over petrol supplies and price stability, led to the removal of the subsidy in 1945.

Without the subsidy it became more difficult for power alcohol to compete with petrol, particularly as production and sales of power alcohol came under ownership of the oil companies (Weir, 1995).

3.2. Coal hydrogenation and Imperial Chemical Industries (ICI)

Developed alongside the Haber–Bosch process for ammonium production in Germany (Hughes, 1969; Smil, 2004), coal hydrogenation via the Bergius Process involved a reaction of hydrogen gas under high-pressure and temperature with creosote (coal-tar) or a creosote and a powdered coal mixture. This yielded a synthetic crude oil. Further distillation of the synthetic oil resulted in a variety of different hydrocarbon oils, and in particular petrol. This was a high quality fuel that could be readily blended with petroleum-derived petrol (Oil Fuel Board, 1931) or with tetra-ethyl lead and iso-octane for a high-octane aviation spirit (Reader, 1977).

The Bergius process became a key method for producing this substitute to petroleum-derived motor fuels in Europe during the inter-war period (see Fig. 1). Between 1927 and 1943, 12 coal hydrogenation plants were built in Germany, two in Britain and one in Korea. The construction of commercial-scale coal hydrogenation plant mostly occurred during the late 1930s when Germany and Britain began rearming and during WWII in an attempt to secure supplies of liquid fuel for the military (National Academy of Sciences, 1977).

The British government played a central role in the early stages of coal hydrogenation development in Britain between 1920 and 1926. From 1927 until 1939, through the Fuel Research Station in Greenwich, the state continued to perform basic research and "scientific study" of the Bergius process. ICI proceeded with commercial development of the technology at the encouragement of the Oil Fuel Board of the Committee of Imperial Defence and the Treatment of Coal Sub-Committee, a sub-committee established in 1929.

ICI formed in late 1926 as an amalgamation of four of Britain's largest chemical firms—Nobel Industries Ltd., British Dyestuffs Corporation, Brunner, Mond & Co., Ltd. (Brunner) and United Alkali Company. ICI's formation was encouraged and brokered by Government officials, partly in response to chemical combines that had emerged in Germany (IG Farben) and the United States (DuPont). Government officials and industrialists held the view that if Britain was to regain its position as industry leader and participate in international trade agreements in the strategically important chemical industry, Britain needed a firm that had similar scale and scope to IG Farben and DuPont (United States Tariff Commission, 1937; Reader, 1977; Pettigrew, 1985).

The vision for ICI was that it would exploit growing and emerging markets within the British Empire, particularly in the agricultural fertiliser sector. Overcapacity of high capital ammonium production equipment used for fertiliser production at ICI's main industrial site in Billingham, Stockton-On-Tees, shortly after its formation in 1926, however, drove the firm to search for new markets. ICI's existing knowledge base in coal hydrogenation due to Brunner's early involvement in the process and the similarities to ammonium production meant that developing petrol-from-coal at Billingham offered ICI a new opportunity for their high capital equipment. With growing concerns about fuel security, declining markets for coal and growing interest in the development of domestic fuel production, ICI built on its knowledge base and began to lobby for state support for a commercial-scale plant at Billingham.

ICI originally hoped coal hydrogenation might be, with government support, a way forward for the stranded assets of the Billingham plant. In the event, however, the process proved to be only cost-competitive with petroleum-derived fuels when oil prices were high and only commercially attractive with a state-guaranteed preference for home-produced hydrocarbon fuels.

3.3. Similarities and differences between the ICI and DCL case-studies

Although clear differences exist between the case studies, discussed further toward the end of this section, both firms shared some similar experiences (Johnson et al., 2014). First, DCL and the chemical firms that merged to form ICI in 1926, played significant roles in WWI, developing a close relationship with the state. Second, both ICI and DCL were 'combines', and were viewed as strategic industries, receiving direct and indirect support from government partly on this basis. Until WWI, alcohol production had not been considered a strategic process but the requirement for munitions alcohol brought government attention. Third, during rearmament in the 1930s, both fuels were recognised for their strategic importance. Finally, ICI and DCL shared a need to identify new markets in the inter-war years because of an excess of supply over demand for their existing products, fertiliser and alcohol. On the supply side, both had surplus capacity developed or planned during WWI, partly through aims to exploit economies of scale and anticipated exports to the British Empire. On the demand side, both experienced growing international competition, as well as falling demand during severe economic downturns.

Whilst ICI and DCL shared experiences, differences did exist. First, whereas a 'preference' (subsidy) for domestic production of hydrocarbon liquid fuels through the British Hydrocarbon Oils Production Act (1934) was orchestrated specifically to assist ICI in the commercial development of an experimental process, DCL received support for an existing product.

Second, the feedstock for ICI's fuel was a home-produced non-renewable fossil fuel, whereas DCL's imported feedstock was renewable. The committee appointed to explore the potential of power alcohol did recognise the advantages of a renewable raw material, but they were in the minority (Committee, 1919). Any rhetoric in support of the development of power alcohol put little emphasis on this property. Third, the petrol-from-coal developed by ICI did not receive any special marketing. A stipulation made in a patent pool agreement (International Hydrogenation Patents agreement) meant that

Table 1
Summary of oil major cartels (adapted from Johnson et al., 2014).

Type of cartel	Name	Description	References
Market-sharing arrangements	Achnacarry Castle Agreement (1928)	• Involved Anglo-Persian, Royal Dutch, and Standard Oil • Aimed to control supply and price at the international level • Agreement strengthened and became more inclusive, particularly following the collapse of oil prices in the economic crisis of 1929–31	Bamberg (1994) Nowell (1994)
	The Combine (early 1920s)	• Involved distribution units of the oil majors in Britain • Aimed to control wholesale and retail prices of motor spirit, prevent price cutting by 'independent' firms, limit new market entrants, standardise motor spirit quality by preventing adulteration of their motor spirit with other, often lower grade fuels • Placed restrictions on retailers permitted to sell its members' motor spirit • Leased/provided finance for petrol pumps, invented in the early 1920s, to secure long-term contracts with retailers for sale of their fuels and gain control over emerging distribution infrastructure • By the mid-1920s, The Combine dominated 85% of the market.	Brunner (1930) Dixon (1963) Fitzgerald (1927)

petrol-from-coal was simply blended with petrol derived from oil and sold as petrol. For DCL, however, power alcohol was branded 'Discol' (later 'Cleveland-Discol') and was promoted as an alcohol fuel made in Britain with superior qualities over petrol. Nevertheless, as oil refining techniques advanced, and with increasing use during the 1920s of tetra-ethyl lead, a cheap fuel-enhancing petrol additive, power alcohol's advantages over petrol were gradually eroded (Weir, 1995).

3.4. Oil major hegemony: fuel distribution and patent pool agreements

The origins and nature of the oil majors are important for the context in which power alcohol and petrol-from-coal emerged. The conditions under which multi-national oil companies developed are well documented and not repeated in detail here (see Refs. Payton-Smith, 1971; Jones 1981; Macbeth, 1985; Bamberg, 1994; Nowell, 1994; Yergin, 2009). In summary, high capital costs of oil distribution and refining infrastructure led to the emergence of 'vertically integrated' firms with global interests in exploration, drilling, production, refining, shipping, distribution, marketing, and related services. In particular, capital intensive distribution networks and associated marketing expertise operated loosely as a 'natural monopoly'. Furthermore, the small number of firms controlling the majority of the world's oil resources was conducive to cartelisation and the exercise of market power. Two types of cartel affected ICI and DCL: market-sharing arrangements and patent pools.

In 1920 the Report on Motor Fuels, an inquiry into high fuel prices, argued that the oil industry and their market-sharing agreements operating at the domestic and international levels (e.g. as in Table 1) had sufficient market power to threaten energy security. The report recommended that the newly established League of Nations act as vehicle for international regulation of the oil market and set maximum and minimum retail prices at the domestic level. It also concluded that domestic production of motor spirits, specifically power alcohol, was ultimately the solution to motor fuel security (Report on Motor Fuel, 1920). Nevertheless, while the state was acutely aware of the restrictive market structure created by the oil cartels, the industry remained unregulated for the duration of the inter-war period (Bridgeman, 1920; Dixon, 1963).

There were several reasons for this. First, the League of Nations was a new institution, and there was no precedent for the regulation of an international firm. Second, whilst government documents are not clear on why recommendations to regulate oil cartels were not approved, a case put forward in 1929 by the oil majors and their distribution subsidiaries offers some insight (Board of Trade, 1929; Economist, 1929). The oil majors framed their argument to government by emphasising that their products were unique in their guaranteed quality, and argued that their parent firms were making substantial infrastructure and oil exploration investments. In contrast, firms such as DCL's distributor, Cleveland, were considered 'independents' and 'pirate' firms, unable or unwilling to make such investments. This perspective was much later confirmed in a report by the Monopolies Commission which implied that in the inter-war period, the state had been keen to attract private investment into the fuel distribution infrastructure (Monopolies Commission, 1965). We suggest that this attitude may have contributed to the state's reluctance to intervene.

Oil majors were also able to affect the economic viability of power alcohol and petrol-from-coal through their involvement in patent pool agreements. These were cartels that developed around specific technologies or processes, often limiting their development. Patent pool agreements existed for both coal hydrogenation (IHP) and power alcohol (ISACo). Little is known

about the ISACo except that it was short-lived as oil companies removed their support early on, choosing instead to focus on fuel enhancing processes, such as additives like tetra-ethyl lead. In contrast, the IHP placed strict restrictions on how the coal hydrogenation technology could be used. Any technological discoveries or modifications to the process by licensees had to be shared with the IHP and only IG Farben was permitted to use the process for purposes other than petrol production. This reduced the economic attractiveness of large-scale production for ICI.

4. State-industry interactions, energy security, the framing of energy and hybrid governance

This section examines how state and industry actors interacted to affect the development of the motor fuels developed by ICI and DCL. Here, we consider government and industry relationships, state support and the effect of oil major hegemony. Through these themes, we illustrate how broader economic, social and technological change and relationships between actors co-produced the development, use and distribution of power alcohol and petrol-from-coal. First, however, we examine the role of energy security as this emerged as a key influence in different ways on all the actors involved.

4.1. Energy security

Concerns about energy security were voiced in the first decade of the 1900s when discussions about the transition of the Royal Navy fleet's fuel from coal to oil began (Jones, 1981). With few domestic crude oil resources, energy security in Britain before and during WWI was mainly framed in terms of securing supply at reasonable prices for the Armed Forces during wartime (Payton-Smith, 1971). A secondary concern, however, discussed in Section 3.4, was the monopolistic organisation of the oil industry and the significant market power it held (Royal Commission on Fuels and Engines, 1913; Jones, 1981).

To guarantee supplies of fuel oil to the Admiralty, in 1914 the state negotiated a controlling interest in the Anglo-Persian Oil Company (APOC). APOC would supply the Admiralty with oil below market price and, during war, the Admiralty would have rights over the entire output. During a time of strong market logic, this level of involvement in a private firm was unprecedented. Nevertheless, while the state had a controlling interest in APOC it agreed not to participate in management activities. The decision to take a controlling interest coincided with an official recognition of fuel oil by the state as a, '*special article*[s] *of warlike material*', exempt from the normal forces of the market (Jones, 1981, p.24).

Experience from WWI shifted state perspectives of energy security during the inter-war period to focus on scarcity and sovereignty, framed in both economic and militaristic terms. First, disruption to Britain's oil supply chains during WWI heightened concerns about resource sovereignty throughout the inter-war period. Second, fears of an absolute petroleum scarcity grew in the 1920s, reaching a climax after publication of a pessimistic report by US President Coolidge's Federal Oil Conservation Board in 1926 (Federal Oil Conservation Board, 1926). Because American oil then dominated the world market (United States Tariff Commission, 1931), the report reinforced the perceived need for British government support of domestic motor fuel production as a hedge against scarcity, particularly during wartime. For example, in Britain, a 1929 report from the Department for Scientific and Industrial Research (DSIR) for the Oil Fuel Board argued that during wartime, domestic, non-petroleum fuels were an attractive proposition, particularly in terms of security of supply (DSIR, 1929).

In the immediate years after the Federal Oil Conservation Board's 1926 report, new oil was found in the US and elsewhere. While oil no longer appeared to be in immediate short supply, Britain experienced two 'oil shocks' in the early 1930s. The first was triggered by a rise in oil prices due to restriction of supply by the oil majors in 1932. This was a coordinated response to the downward pressures on prices during the depression (Reader, 1977; Stranges, 1984). The second 'shock' was triggered by the collapse of the D'Arcy concession in 1933. The concession was a controversial 60-year agreement (from 1909) between Iran and APOC, which granted APOC exclusive oil exploration rights to a particular area of Iran (Oil Fuel Board, 1933; Kakhki, 2008). After hurried negotiations a new agreement was reached. Nevertheless, the experience brought into relief Britain's dependence on foreign oil.

Both case-studies provide evidence that energy security played a significant role in the framing of energy, and the availability, shaping and timing of state support for domestic production of motor spirits. Three issues in particular: perceptions of fuel scarcity, dependence on non-domestic, non-colonial sources of fuel, and Britain's balance of payments, highlighted the economic and military potential of alternative liquid fuels to reduce Britain's dependence on imports. Consequently, a multi-faceted concept of energy security emerged and had become a key concern, providing impetus for research into, and support for, the domestic production of motor fuel.

Both ICI and DCL framed their lobbying accordingly. ICI, in its negotiations with the Board of Trade for a state subsidy, consistently highlighted the value of Britain's indigenous coal supply for energy independence during peace and wartime, and its impact on Britain's trade deficit (Reader, 1977). Drawing issues of sovereignty together with the availability of raw materials, DCL emphasised that the main raw material for power alcohol (molasses) was widely available within the Empire.

For DCL, post-WWI oil shortage fears provided the underlying rationale for government to extend the allowance on industrial alcohol to power alcohol (from 1921), encouraging production (Yergin, 2009). Even when oil became more readily available, scarcity of supply concerns, combined with the two 'oil price shocks' eventually led to the passing of the British Hydrocarbon Oils Production Act in 1934, encouraging ICI to operationalise its plans for an experimental industrial-scale coal hydrogenation plant based on its Billingham assets.

4.2. Government and industry relationships and their influence on governance

Both World Wars had significant influence on DCL's and ICI's relationships with government. Due to the shortages of essential supplies during WWI, supply contracts with government established close relationships with state actors and institutions. Government and industry relationships and mutual understanding also benefitted from the sharing of personnel during wartime.

In particular, multiple government departments interacted with industry to address munitions concerns. In 1917, for example, the Advisory Committee on Alcohol Supplies, in search of alcohol and acetone supplies for munitions, established a pattern of interaction between distillers and government over alcohol supplies that continued throughout the inter-war period and WWII (Weir, 1995). Similarly, during WWI chemical firm Brunner worked closely with the Ministry of Munitions through manufacture and research contracts (Reader, 1975). During the inter-war years, the Admiralty and the Air Ministry also played a considerable role in RD&D of alternative fuels, working closely with ICI and DCL.

Moreover, the wartime economy experiences had changed perspectives on the virtues of co-operation, mergers, and large-scale enterprise (Hannah, 1983), with government supporting business amalgamation (Broadberry, 1986). The size and scope of ICI's and DCL's businesses, born out of the second industrial revolution and intensified through the 'amalgamation movement', gave them the capability to communicate with, and influence, government and industry.

The amalgamation movement more generally transformed industrial organisation and required experienced managers, who were often recruited into industry from the civil service. For example, W.H. Coates and Josiah Stamp were both senior civil servants from the Inland Revenue recruited into senior ICI management positions early in its history. This, combined with an established framework of cooperation developed during wartime, helped create a 'revolving door', giving DCL and ICI opportunities to influence government policy at the highest level and an appreciation of how to exert that influence (Weir, 1995; Reader, 1977).

4.3. Indirect state support and shifts in governance logics

The wartime economy of WWI and economic instability during the inter-war years contributed to shifts in governance logics. For example, during WWI there was a change from traditional hard-line laissez-faire approaches to acceptance of greater state intervention. Despite a rapid return to free-trade ideals following WWI, however, government continued to support industries that had been strategically important during the war.

Before WWI, state intervention in industry was limited in terms of sharing knowledge and resources (Edgerton and Horrocks, 1994). However, during WWI and the inter-war period, state engagement with, and support for, scientific research and strategic defence industries was seen as necessary and urgent. Furthermore, received wisdom about the emerging heavy chemical industry was that Britain was falling behind Germany and the US (Haber, 1971; Reader, 1977). During WWI severe criticisms of Britain's flagging technological prowess were voiced by high profile political figures, who looked to Germany's chemical industry as an exemplar due to its rapid transformation into a world leader (Macleod and Andrews, 1969; Marsch, 2000).

Both ICI, with its capacity and experience for producing chemicals with military uses, and DCL, with its capacity for producing large quantities of alcohol, useful for munitions, were considered strategic industries. In addition, DCL's ethyl alcohol was an essential raw material for the chemical industry, the development of which was one of the rationales for ICI's creation (Weir, 1995). For these reasons, both ICI and DCL became a focus of government attention.

From 1914 the state took a central role in planning, financing, and directing the activities of strategically important manufacturing firms (Hannah, 1983). This led, in 1915, to the formation of the Committee for Scientific and Industrial Research, charged with promoting research in industry and education. It became the Department for Scientific Industrial Research (DSIR) in 1916.

The DSIR played a key role in indirect state support for ICI and DCL's motor spirits in two ways. First, its preferred method was to support industrialists in highly specialised sectors including the chemical industry, to form research associations subsidised by government funds. For example, two DSIR grants to the British Colliery Owners' Research Association in the early 1920s provided funds for academic research into coal hydrogenation, the earliest evidence of state support for this emerging technology (Stranges, 1984).

Second, the DSIR itself undertook research considered to be of national importance and deemed unsuitable for private enterprise (Sayer, 1950). As DCL's power alcohol and ICI's petrol-from-coal were both considered strategically important, the fuels received support in this way. The DSIR supported power alcohol by arranging funding for, and initiating, the testing of power alcohol in a fleet of London's Omnibuses (The Times, 1920).[1] Additionally, it provided finance for a group of private investors to acquire patent licences for coal hydrogenation (Stranges, 1984). The DSIR also formed several Departmental research stations, including the Fuel Research Station (FRS), established in 1919. The FRS played a significant role in developing coal hydrogenation processes and the testing of both fuels.

[1] In a modern equivalent, in 2003–05 27 hydrogen-powered, fuel cell buses were placed in the public transport fleets of nine European cities, including London (http://ec.europa.eu/energy/res/fp6_projects/doc/hydrogen/deliverables/summary.pdf, accessed 23/05/14).

Establishment of the DSIR and FRS demonstrated governmental commitment to investigating the promise of alternative fuels, underpinned by the notion of their potential strategic importance. Both organisations worked closely with ICI and DCL.

4.4. Liquid fuels and the emergence of hybrid governance

Our case studies illustrate that motives for DCL and ICI's engagement with large-scale production of alternative liquid fuels and government support for them were dynamic and diverse. As noted, both ICI and DCL shared a similar need to identify new markets in the inter-war years because of an excess of supply over demand. Government motives included attempts to maintain British leadership in the synthetic chemical industry in a competitive international market in the early 1920s, the desire to find ways of modernising and identifying new markets for a domestic coal industry in the mid-1920s, and a potential means of rebalancing trade deficits to stabilise public debt. Added to this was the recession in the early-1930s and rearmament as geopolitical pressures grew toward the middle of the decade. All these factors contributed to the emergence of a multi-faceted case for supporting the production of domestic fuels. This was reinforced by the growing acceptability of government intervention in strategic industries during and after WWI.

Fuel security concerns, in particular, led the government to adopt several sometimes conflicting approaches in order to secure fuel supplies: maintaining a controlling interest in APOC, minimising state intervention in the oil industry in terms of regulation, whilst providing subsidies for the development and production of alternative liquid fuels.

Liquid fuels first emerged outside their military uses as a 'luxury commodity' or 'article of general consumption'. From 1914 onwards, as a consequence of concerns relating to security of supply and cost, significance to the defence services and post-WWI diffusion through society, liquid fuels became privileged as a 'special article of warlike material' (Jones, 1981) and/or a 'social service', in the manner described in Section 2. This shift in the framing of liquid fuels led the government to work closely with the oil majors, specifically APOC, to achieve a secure fuel supply. Government expectations that the oil industry would provide the necessary investment into distribution infrastructure and continue with oil exploration, however, contributed to a willingness to accept the market power of the domestic level cartel, The Combine, and an unwillingness to intervene through regulation.

Oil company ownership and control over oil resources and distribution networks effectively limited the state's agency to control oil prices and willingness to influence fuel distribution. During periods of heightened energy insecurity, the state instead legitimised the expense of supporting alternative fuels by means of a subsidy. Whilst subsidies issued by the Board of Trade were provided to support DCL's and ICI's emerging fuels, oil major control of fuel distribution hindered the likelihood of their success. The Oil Fuel Board and other commentators such as the Economist recognised this inconsistency and the potential hostility of the oil majors to alternative fuels (Oil Fuel Board, 1930; Economist, 1927). Non-interference with the oil majors' control of fuel distribution was, however, implicitly endorsed by the government. This effectively amounted to a 'hybrid' governance logic, with state intervention through subsidies on the one hand, and an unwillingness to regulate the oil majors on the other. As such, DCL and ICI's experience was of a policy environment in which state subsidisation made both fuels cost-competitive with imported motor fuels, but offered no support for access to markets through The Combine-controlled distribution infrastructure, or protection from its drive to exclude alternative fuels from the market.

From 1937 onwards, reviews into future plans for domestically produced substitute fuels (Rivett, 1937; Falmouth Committee, 1938 for ICI; May Committee, 1945 for DCL), considered that despite direct subsidies, they remained costly and uncompetitive and this was unlikely to change. Furthermore, as oil supplies became more certain after WWII, the government's commitment to securing energy supplies favoured oil company developments to a greater extent over alternative domestic fuel sources and state subsidies were removed (in 1945 for power alcohol and 1958 for domestically produced hydrocarbon oils).

The inter-war period therefore represented a period of dynamic governance in which an unstable combination, or hybrid, of market- and state-led logics existed. This reflected the political complexity of the period and the dynamic interaction of energy security, framing and governance.

5. Conclusions and insights from a failed transition

We suggest that, with the aid of the analytical approach set out in Section 2 and Fig. 3, our case studies have enabled the drawing of a rich and complex picture of how two different large firms' attempts at path creation through the introduction of new fuels to rival the incumbent petrol in an unstable economic context, interacted with varying perceptions of energy security, changes in the framing of energy and shifts in governance. The paper has illustrated that there were multiple influences on the development of alternative fuels during the inter-war period, which informed and influenced government and industry interactions and responses in different but often mutually compatible, ways. The state, variously wanting to support the coal industry, ensure domestic supplies of military matériel (chemicals and alcohol), and to address concerns about unstable crude oil markets and fluctuations in the petrol price, provided both direct and indirect support for these fuels. Despite this support, however, both power alcohol and petrol-from-coal depended on specialised complementary assets (viz. fuel distribution networks) owned by international oil companies. Whilst government recognised the monopoly position of these incumbents in the control of distribution infrastructure, their concerns for secure oil supplies problematised regulation of access to it, hindering the market penetration of these economically challenged alternatives. In the final analysis,

government effectively gave priority to the petroleum industry, while to some extent continuing to hedge their bets by also supporting these alternative fuels.

Over the inter-war period the increasing civil and military dependence of state and society on liquid fuels meant that they gradually became more of a necessity than a luxury commodity. Also, because of the anxieties of government for secure fuel supplies, and the perceived capacity of oil companies to provide this, liquid fuels were subject to a fluctuating, hybrid form of governance throughout the period. During times of heightened energy security concerns, liquid fuels were seen more as a 'special article' or 'social service' rather than a 'market commodity', and subject to greater state intervention. In contrast, during times when energy was more secure, there was less state intervention and market conditions dominated. The inter-war period therefore represented a period of dynamic, shifting governance in which combinations, or hybrids, of market- and state-led logics existed.

The observation of these dynamics in this paper came about through the application of a relational approach, informed by our unpacking of the interactions between perceptions of energy security, the framing of energy and unstable governance logics. This enabled us to observe not only how government and market logics co-evolved in a pliable action space but also how shifting modes of governance and actor agency affected energy decision-making, facilitating the capture of diverse characteristics of the failed-transition to the two alternative fuels.

We argue that insights from these historical case-studies can be useful not only for understanding the past but also for interpreting the contemporary dynamics of energy policy and sustainability transitions, even though so much has changed in the UK and elsewhere. For example, these cases offer a partial historical analogue for current situations which have shown growing emphasis on the hybrid role of state and market in energy governance and concern with whether incumbent actors through their ownership of technologies and infrastructures will reinforce or constrain low carbon energy transitions.

Thus, the UK government's recent electricity market reform initiatives (DECC, 2015) form part of a movement away from the UK's dominant emphasis (after the late 1980s and 1990s energy market liberalisation) on the role of the market and the private sector in delivering narrowly defined energy policy objectives that put little emphasis on wider social and environmental policy concerns and somewhat downplayed aspects of longer term energy security (Pearson and Watson, 2011). The government has moved towards a more hybrid form of governance that aims to address these wider objectives and respond to the thrust of European Union energy policy on renewable energy (Emamian, 2014; International Energy Agency, 2014). The cases resonate with how changing concerns and priorities within the 'trilemma' of energy and climate policy objectives (climate change and environment, energy security, and affordability), particularly in relation to energy security, have interacted with the framing and governance of energy and energy policy and its instruments (Bolton and Foxon, 2013; Mitchell et al., 2013; DECC, 2015). Moreover, the UK has not been alone in introducing a variety of support mechanisms for energy security and low carbon fuels, including renewables, that reflect attempts to meet national and international policy objectives and resolve conflicts between them (International Energy Agency, 2014).

There are significant current concerns internationally over the funding, availability of and access to distribution infrastructure for refuelling vehicles with alternative fuels, whether with bio-based fuels, hydrogen or electricity. This is despite the increased availability of flex-fuel engines for military uses and their successful adoption for non-military uses in some countries that have encouraged them (as in the recent growing ownership of engines that can switch between ethanol and petrol in Brazil and refuel at filling stations that had been required to dispense both fuels (Macedo, 2007)—a case of the successful creation of alternative fuel distribution infrastructure). There is also a broader issue of 'infrastructure governance' in energy transitions because of the technological and economic complementarities of infrastructures with many technologies and sectors, such that they form systems that can powerfully constrain or enable significant change (Teece, 1986; Jonsson, 2000; Helm, 2009; Markard, 2011; Andersen, 2014). In the UK low carbon energy sources are being advanced as alternatives to fossil-based transport fuels, and low carbon technologies are proposed as alternatives to natural gas based heating systems (Chaudry et al., 2015). In situations like these, the case studies offer examples of how to unpick the complexities and pitfalls involved, including issues of policy inconsistency at a point in time and over time and of the availability of and access to distribution or other complementary assets. Our study clearly illustrates the challenges faced by emerging technological substitutes that may be reliant on initial policy support and access to complementary assets owned and/or controlled by incumbents.

Whilst most studies explore successful transitions, this study casts light on how and why attempts at path creation through innovation can founder and potential transitions fail in periods of shifting governance and tensions between incumbents and newcomers. We suggest that, while our case studies do not offer any blueprint for policy making, decision makers could benefit from an awareness of the interactions between governance logics, the framing of energy and energy security, the behaviours of incumbents and the challenges of path creation, of the kind described here. Policy makers could usefully reflect on this story of a failed transition.

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